KERALA DIPLOMA CURRICULUM REVISION 2026

Course 2052 | II | As specified in the official PDF | Source-grounded

Fundamentals of Internal Combustion Engines

bearings and flywheel.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 2052 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 2052.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma Programme
Course code	2052
Course title	Fundamentals of Internal Combustion Engines
Semester	II
Course category	As specified in the official PDF

Item	Official information
Periods per week	As specified in the official PDF
Periods per semester	As specified in the official PDF
Credits	As specified in the official PDF
Prerequisite	Topic Course code Course Title Semester Classification of IC engine, Operation of two-stroke and four-stroke internal 1051 Basic Automobile Engineering I combustion engines Course Outcome CO Course Outcome Blooms Taxonomy Level CO1 Explain construction, materials and manufacturing methods of engine components. Understand CO2 Explain working principles and interaction of reciprocating and rotating engine components. Understand CO3 Explain valve mechanisms, cylinder head types and modern valve timing systems used in IC engines. Understand Explain and compare lubrication and cooling systems considering efficiency, reliability, maintenance and CO4 Understand environmental aspects of engine operation. CO-PO mapping COURSE ARTICULATION MATRIX PROGRAM OUTCOMES
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

21 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- bearings and flywheel.

Course outcomes

CO	Official outcome	Study level
CO1	3 2 - - - - - - - - 2	See official syllabus
CO2	3 2 2 - - 2 - - - - 2	See official syllabus
CO3	3 2 2 2 2 2 2 - 2 2 2	See official syllabus
CO4	3 2 - 2 - 2 2 - - - 2 Total 12 8 4 4 2 6 4 - 2 2 8 Correlation Level 3 2 2 2 2 2 2 - 2 2 2 Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - “-” Teaching and Assessment Scheme Diploma Curriculum Revision 2026 2052 Page #1 Examination Scheme Teaching Scheme/Week Self-learning Total Credits Theory Practical (SS)/Week Hours/Semester C Total L T P S CIE ESE Total CIE ESE Total 4 0 0 0 6 60 4 40 60 100 0 0 0 100 L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One	See official syllabus

CO	Official outcome	Study level
	unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination Syllabus - Major Topics Instructional Module Title Major Topics Hours L T Introduction to engine construction; cylinder block and crankcase, cylinder head, oil Engine Construction pan, crank case ventilation and its role in emission control, intake and exhaust	

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3 2 - - - - - - - - 2
CO2	CO2 3 2 2 - - 2 - - - - 2
CO3	CO3 3 2 2 2 2 2 2 - 2 2 2
CO4	CO4 3 2 - 2 - 2 2 - - - 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	15 0	3	As specified
Module 2	Camshaft (construction and materials), Camshaft drives, Vibration dampers, 15 0	6	As specified
Module 3	Camshaft arrangements (SOHC & DOHC), Valve timing diagram of four-stroke engine 15 0	5	As specified
Module 4	filtering systems, Oil cooler and pressure relief valve, Engine Cooling systems (air and 15 0	7	As specified

MODULE 1

15 0

This module is presented from the official syllabus scope for Fundamentals of Internal Combustion Engines. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: 15 0
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

Details - I manifolds, gaskets, cylinder liners (dry and wet), piston (construction, materials and

Details - I manifolds, gaskets, cylinder liners (dry and wet), piston (construction, materials and is an official topic in Module 1 of Fundamentals of Internal Combustion Engines. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Details - I manifolds, gaskets, cylinder liners (dry and wet), piston (construction, materials and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, details - i manifolds, gaskets, cylinder liners (dry and wet), piston (construction, materials and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Details - I manifolds, gaskets, cylinder liners (dry and wet), piston (construction, materials and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Details - I manifolds, gaskets, cylinder liners (dry and wet), piston (construction, materials and □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

types), piston rings.

types), piston rings. is an official topic in Module 1 of Fundamentals of Internal Combustion Engines. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify types), piston rings. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, types), piston rings. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What types), piston rings. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] types), piston rings. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Engine Construction

Engine Construction is an official topic in Module 1 of Fundamentals of Internal Combustion Engines. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Engine Construction using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, engine construction should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Engine Construction means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Engine Construction □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
15         0
              Details - I             manifolds, gaskets, cylinder liners (dry and
wet), piston (construction, materials and
                                      types), piston rings.
                                      Connecting rod, piston pin, Crankshaft
(construction and materials), Engine Bearings,
              Engine Construction
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Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Camshaft (construction and materials), Camshaft drives, Vibration dampers, 15 0

This module is presented from the official syllabus scope for Fundamentals of Internal Combustion Engines. Use the topic cards for learning and the source transcript for exact coverage.

As specified official hours	**CO-linked** see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Camshaft (construction and materials), Camshaft drives, Vibration dampers, 15 0
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

Camshaft (construction and materials), Camshaft drives, Vibration dampers, 15 0

Camshaft (construction and materials), Camshaft drives, Vibration dampers, 15 0 is an official topic in Module 2 of Fundamentals of Internal Combustion Engines. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Camshaft (construction and materials), Camshaft drives, Vibration dampers, 15 0 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, camshaft (construction and materials), camshaft drives, vibration dampers, 15 0 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Camshaft (construction and materials), Camshaft drives, Vibration dampers, 15 0 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Camshaft (construction and materials), Camshaft drives, Vibration dampers, 15 0 □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Details - II

Details - II is an official topic in Module 2 of Fundamentals of Internal Combustion Engines. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Details - II using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, details - ii should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Details - II means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Details - II □□□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□□.

Flywheel, Types of Flywheel.

Flywheel, Types of Flywheel. is an official topic in Module 2 of Fundamentals of Internal Combustion Engines. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Flywheel, Types of Flywheel. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, flywheel, types of flywheel. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Flywheel, Types of Flywheel. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] Flywheel, Types of Flywheel. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Types of Cylinder Heads (L, T, F, I Heads), Poppet valve (construction and materials),

Types of Cylinder Heads (L, T, F, I Heads), Poppet valve (construction and materials), is an official topic in Module 2 of Fundamentals of Internal Combustion Engines. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Types of Cylinder Heads (L, T, F, I Heads), Poppet valve (construction and materials), using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, types of cylinder heads (l, t, f, i heads), poppet valve (construction and materials), should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Types of Cylinder Heads (L, T, F, I Heads), Poppet valve (construction and materials), means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Types of Cylinder Heads (L, T, F, I Heads), Poppet valve (construction and materials), □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Sodium cooled valves, valve train components, Valve actuating mechanisms,

Sodium cooled valves, valve train components, Valve actuating mechanisms, is an official topic in Module 2 of Fundamentals of Internal Combustion Engines. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Sodium cooled valves, valve train components, Valve actuating mechanisms, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, sodium cooled valves, valve train components, valve actuating mechanisms, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Sodium cooled valves, valve train components, Valve actuating mechanisms, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Sodium cooled valves, valve train components, Valve actuating mechanisms, □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Types of cylinder head

Types of cylinder head is an official topic in Module 2 of Fundamentals of Internal Combustion Engines. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Types of cylinder head using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, types of cylinder head should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Types of cylinder head means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Types of cylinder head □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Camshaft (construction and materials), Camshaft drives, Vibration dampers,
15         0
              Details - II
                                      Flywheel, Types of Flywheel.
                                  Types of Cylinder Heads (L, T, F, I Heads),
Poppet valve (construction and materials),
                                  Sodium cooled valves, valve train components,
Valve actuating mechanisms,
           Types of cylinder head
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Camshaft arrangements (SOHC & DOHC), Valve timing diagram of four-stroke engine 15 0

This module is presented from the official syllabus scope for Fundamentals of Internal Combustion Engines. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Camshaft arrangements (SOHC & DOHC), Valve timing diagram of four-stroke engine 15 0
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

Camshaft arrangements (SOHC & DOHC), Valve timing diagram of four-stroke engine 15 0

Camshaft arrangements (SOHC & DOHC), Valve timing diagram of four-stroke engine 15 0 is an official topic in Module 3 of Fundamentals of Internal Combustion Engines. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Camshaft arrangements (SOHC & DOHC), Valve timing diagram of four-stroke engine 15 0 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, camshaft arrangements (sohc & dohc), valve timing diagram of four-stroke engine 15 0 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Camshaft arrangements (SOHC & DOHC), Valve timing diagram of four-stroke engine 15 0 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Camshaft arrangements (SOHC & DOHC), Valve timing diagram of four-stroke engine 15 0 □□□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

and valve mechanism

and valve mechanism is an official topic in Module 3 of Fundamentals of Internal Combustion Engines. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify and valve mechanism using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, and valve mechanism should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What and valve mechanism means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ and valve mechanism □□□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

(theoretical and actual), Variable Valve Timing (VVT) – Concept, Need and Types of

(theoretical and actual), Variable Valve Timing (VVT) – Concept, Need and Types of is an official topic in Module 3 of Fundamentals of Internal Combustion Engines. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify (theoretical and actual), Variable Valve Timing (VVT) – Concept, Need and Types of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, (theoretical and actual), variable valve timing (vvt) – concept, need and types of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What (theoretical and actual), Variable Valve Timing (VVT) – Concept, Need and Types of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ (theoretical and actual), Variable Valve Timing (VVT) – Concept, Need and Types of □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

multi-grade oils, Oil additives, Engine lubrication systems, Types of oil pumps, Oil

multi-grade oils, Oil additives, Engine lubrication systems, Types of oil pumps, Oil is an official topic in Module 3 of Fundamentals of Internal Combustion Engines. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify multi-grade oils, Oil additives, Engine lubrication systems, Types of oil pumps, Oil using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, multi-grade oils, oil additives, engine lubrication systems, types of oil pumps, oil should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What multi-grade oils, Oil additives, Engine lubrication systems, Types of oil pumps, Oil means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] multi-grade oils, Oil additives, Engine lubrication systems, Types of oil pumps, Oil [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Automobile Engine

Automobile Engine is an official topic in Module 3 of Fundamentals of Internal Combustion Engines. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Automobile Engine using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, automobile engine should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Automobile Engine means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Automobile Engine □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Camshaft arrangements (SOHC & DOHC), Valve timing diagram of four-stroke engine
15          0
           and valve mechanism
                                  (theoretical and actual), Variable Valve Timing
(VVT) – Concept, Need and Types of
                                  VVT.
                                      Concept and methods of lubrication,
properties of lubricating oil, single-grade and
                                      multi-grade oils, Oil additives, Engine
lubrication systems, Types of oil pumps, Oil
          Automobile Engine
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

filtering systems, Oil cooler and pressure relief valve, Engine Cooling systems (air and 15 0

This module is presented from the official syllabus scope for Fundamentals of Internal Combustion Engines. Use the topic cards for learning and the source transcript for exact coverage.

As specified official hours	**CO-linked** see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: filtering systems, Oil cooler and pressure relief valve, Engine Cooling systems (air and 15 0
- Official duration: As specified
- Official topic entries captured: 7
- The full source excerpt is preserved below for coverage checking.

filtering systems, Oil cooler and pressure relief valve, Engine Cooling systems (air and 15 0

filtering systems, Oil cooler and pressure relief valve, Engine Cooling systems (air and 15 0 is an official topic in Module 4 of Fundamentals of Internal Combustion Engines. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify filtering systems, Oil cooler and pressure relief valve, Engine Cooling systems (air and 15 0 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, filtering systems, oil cooler and pressure relief valve, engine cooling systems (air and 15 0 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What filtering systems, Oil cooler and pressure relief valve, Engine Cooling systems (air and 15 0 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] filtering systems, Oil cooler and pressure relief valve, Engine Cooling systems (air and 15 0 [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

water cooling), Cooling system components, Radiator cap, Expansion reservoir, and

water cooling), Cooling system components, Radiator cap, Expansion reservoir, and is an official topic in Module 4 of Fundamentals of Internal Combustion Engines. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify water cooling), Cooling system components, Radiator cap, Expansion reservoir, and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, water cooling), cooling system components, radiator cap, expansion reservoir, and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What water cooling), Cooling system components, Radiator cap, Expansion reservoir, and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ water cooling), Cooling system components, Radiator cap, Expansion reservoir, and □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

antifreeze solutions.

antifreeze solutions. is an official topic in Module 4 of Fundamentals of Internal Combustion Engines. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify antifreeze solutions. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, antifreeze solutions. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What antifreeze solutions. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] antifreeze solutions. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Detailed Syllabus

Detailed Syllabus is an official topic in Module 4 of Fundamentals of Internal Combustion Engines. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Detailed Syllabus using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, detailed syllabus should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Detailed Syllabus means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Detailed Syllabus [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Mode of

Mode of is an official topic in Module 4 of Fundamentals of Internal Combustion Engines. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Mode of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, mode of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Mode of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Mode of [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Mapped Instructional

Mapped Instructional is an official topic in Module 4 of Fundamentals of Internal Combustion Engines. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Mapped Instructional using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, mapped instructional should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Mapped Instructional means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Mapped Instructional [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Subtopic Student Learning Outcome delivery Taxonomy Level

Subtopic Student Learning Outcome delivery Taxonomy Level is an official topic in Module 4 of Fundamentals of Internal Combustion Engines. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Subtopic Student Learning Outcome delivery Taxonomy Level using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, subtopic student learning outcome delivery taxonomy level should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Subtopic Student Learning Outcome delivery Taxonomy Level means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-▯▯ Subtopic Student Learning Outcome delivery Taxonomy Level ▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯ definition/meaning ▯▯▯▯▯▯▯▯▯▯▯▯▯. ▯▯▯▯▯▯ ▯▯▯▯▯▯ ▯▯▯▯▯▯▯, steps, application ▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯. Technical terms English-▯ ▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯▯.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
filtering systems, Oil cooler and pressure relief valve, Engine Cooling systems (air
and           15         0
          Cooling & Lubrication
                                      water cooling), Cooling system components,
Radiator cap, Expansion reservoir, and
                                      antifreeze solutions.

Detailed Syllabus

Mode of

Mapped                          Instructional
          Subtopic                           Student Learning Outcome
delivery             Taxonomy Level

CO                                Hours

(L/T)
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

- for Self-Learning
- Week Self-Learning description Hours Module I
- Study manufacturing methods of cylinder block and cylinder head and analyze their effect on engine performance, cost and
- sustainability.
- Prepare comparison chart of dry and wet cylinder liners considering performance, maintenance requirements and
- environmental impact.
- Sketch and classify different types of pistons and explain their applications in automobile engines. 5
- Case study on aluminium vs cast iron engine block construction considering strength, weight, manufacturing cost and
- sustainability. Module II
- Draw crankshaft layout showing journals and crankpins and explain its importance for engine reliability and safe operation. 5
- Study bearing clearance and oil film formation in engine bearings. 4
- Compare types of flywheel with applications considering durability, vibration control and maintenance aspects. 4
- Analyze auxiliary drive systems in engines and discuss their role in improving vehicle performance, safety and user comfort. 5
- Compare gear, chain and belt driven camshaft systems considering reliability, noise, maintenance and cost. 5 Module III
- Draw theoretical and actual valve timing diagrams and analyze their influence on engine performance. 5
- Compare L, T, F and I head engines considering design features and application in automobiles. 4
- Prepare comparison chart of SOHC and DOHC systems considering performance, complexity, maintenance and cost. 4
- Study Variable Valve Timing (VVT) systems and evaluate their role in improving engine efficiency and reducing emissions. 4
- Prepare a sketch or simple model of poppet valve assembly and present its working through a small group discussion. 4 Module IV
- Compare splash and pressure lubrication systems considering efficiency, reliability and environmental aspects. 4

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
– Theory
                                                 CA1                   CA2
CA3                     CA4                     CA5            CA6                  ESE

       Mode            Attendance             Self-learning     Self-learning
Self-learning          Self-learning              Series      Series
Written
                                               activities        activities
activities             activities               Exam        Exam
Exam
                                                (Module 1)        (Module 2)
(Module 3)             (Module 4)             (2 modules) (2 modules)
(theory)
   Duration
2 hours          2 hours           2.5 hours
  Exam mark                                         15                  15
15                          15                 50              50                  60
 Converted to
20               20
       Marks                5                                                        15
20                          60
       Total
40                                                                                      60
   Tentative              End of
End of
                                              By 4th week       By 7th week
By 11th week           By 14th week             8th week         15th week
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Details - I manifolds, gaskets, cylinder liners (dry and wet), piston (construction, materials and. (3 marks)

Answer: Begin with the standard definition or identity of *Details - I manifolds, gaskets, cylinder liners (dry and wet), piston (construction, materials and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain types), piston rings.. (3 marks)

Answer: Begin with the standard definition or identity of *types), piston rings.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Engine Construction. (3 marks)

Answer: Begin with the standard definition or identity of *Engine Construction*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain Camshaft (construction and materials), Camshaft drives, Vibration dampers, 15 0. (3 marks)

Answer: Begin with the standard definition or identity of *Camshaft (construction and materials), Camshaft drives, Vibration dampers, 15 0*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Details - II. (3 marks)

Answer: Begin with the standard definition or identity of *Details - II*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Flywheel, Types of Flywheel.. (3 marks)

Answer: Begin with the standard definition or identity of *Flywheel, Types of Flywheel.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Types of Cylinder Heads (L, T, F, I Heads), Poppet valve (construction and materials),. (3 marks)

Answer: Begin with the standard definition or identity of *Types of Cylinder Heads (L, T, F, I Heads), Poppet valve (construction and materials),*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 3

Define or explain Camshaft arrangements (SOHC & DOHC), Valve timing diagram of four-stroke engine 15 0. (3 marks)

Answer: Begin with the standard definition or identity of *Camshaft arrangements (SOHC & DOHC), Valve timing diagram of four-stroke engine 15 0*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain and valve mechanism. (3 marks)

Answer: Begin with the standard definition or identity of *and valve mechanism*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain (theoretical and actual), Variable Valve Timing (VVT) – Concept, Need and Types of. (3 marks)

Answer: Begin with the standard definition or identity of *(theoretical and actual), Variable Valve Timing (VVT) – Concept, Need and Types of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain multi-grade oils, Oil additives, Engine lubrication systems, Types of oil pumps, Oil. (3 marks)

Answer: Begin with the standard definition or identity of *multi-grade oils, Oil additives, Engine lubrication systems, Types of oil pumps, Oil*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Module 4

Define or explain filtering systems, Oil cooler and pressure relief valve, Engine Cooling systems (air and 15 0. (3 marks)

Answer: Begin with the standard definition or identity of *filtering systems, Oil cooler and pressure relief valve, Engine Cooling systems (air and 15 0*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain water cooling), Cooling system components, Radiator cap, Expansion reservoir, and. (3 marks)

Answer: Begin with the standard definition or identity of *water cooling), Cooling system components, Radiator cap, Expansion reservoir, and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps,

□□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain antifreeze solutions.. (3 marks)

Answer: Begin with the standard definition or identity of *antifreeze solutions.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Detailed Syllabus. (3 marks)

Answer: Begin with the standard definition or identity of *Detailed Syllabus*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Details - I manifolds, gaskets, cylinder liners (dry and wet), piston (construction, materials and**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Camshaft (construction and materials), Camshaft drives, Vibration dampers, 15 0**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Camshaft arrangements (SOHC & DOHC), Valve timing diagram of four-stroke engine 15 0**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **filtering systems, Oil cooler and pressure relief valve, Engine Cooling systems (air and 15 0**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- URL Modules Covered
- https://nptel.ac.in/courses/112/104/112104033 All

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTR syllabus PDF for Course 2052. Verify institutional updates against the current official curriculum.